

A MINI PROJECT

on

**Performance Evaluation of 5G OTFS systems under
realistic fading channels**

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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(NAAC Accredited – A Grade | NBA Accredited B. Tech. (EEE, ECE and CSE))
Bachupally, Hyderabad – 500090**

2025-26

DECLARATION

We hereby declare that the work described in this report, entitled “**Performance Evaluation of 5G OTFS systems under realistic fading channels**” which is being submitted by us in partial fulfilment for the award of the degree of **Bachelor of Technology** in the **Department of Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women (Autonomous)**, affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad – 500090 is the result of original work carried out by us under the guidance of **Mr. Saikumar Tara, Assistant Professor**.

This work has not been submitted for any Degree/Diploma of this or any other institute/university to the best of our knowledge and belief.

Place: Hyderabad

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Certificate

This is to certify that the major/mini project report, entitled “**Performance Evaluation of 5G OTFS systems under realistic fading channels**” is a record of bonafide work carried out by **S. Cynthana (23WH1A0419), G. Saranya (23WH1A0439), T. Sanjana (23WH1A0416), Shahnaz (23WH1A0418)** in partial fulfilment for the award of the degree of **Bachelor of Technology** in the department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women (Autonomous)**, affiliated to **Jawaharlal Nehru Technological University Hyderabad, Kukatpally, Hyderabad – 500090**.

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ABSTRACT

Modern wireless communication systems require high data rates, low latency, and reliable performance under dynamic channel conditions. Orthogonal Frequency Division Multiplexing (OFDM), widely used in current systems, offers high spectral efficiency but suffers from performance degradation in high mobility environments due to Doppler-induced inter-carrier interference.

To overcome these limitations, Orthogonal Time Frequency Space (OTFS) modulation has been proposed, which operates in the delay-Doppler domain and provides improved robustness against time-varying channels by spreading information across the entire time-frequency grid.

In this project, a comparative analysis of OFDM and OTFS systems is carried out under various fading channel conditions, including AWGN, Rayleigh, Rician, and Weibull channels. The systems are implemented using MATLAB, and performance is evaluated in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), Doppler impact, and spectral efficiency.

Simulation results show that while OFDM performs well under low mobility conditions, OTFS significantly outperforms OFDM in high mobility and multipath environments. This study highlights the effectiveness of OTFS as a promising modulation technique for future wireless communication systems.

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1. INTRODUCTION

1.1 Overview of the Project

Wireless communication systems have evolved significantly from 4G to 5G and are now progressing towards 6G technologies. These systems require high data rates, low latency, and reliable communication under dynamic conditions. One of the most widely used modulation techniques is Orthogonal Frequency Division Multiplexing (OFDM), which divides a high-rate data stream into multiple low-rate parallel subcarriers.

Despite its advantages, OFDM suffers from performance degradation in high mobility environments due to Doppler effects and time-varying channels. To overcome these limitations, Orthogonal Time Frequency Space (OTFS) modulation has been introduced, which operates in the delay-Doppler domain instead of the conventional time-frequency domain.

In this project, a detailed comparison between OFDM and OTFS is performed using MATLAB simulation. The system evaluates performance across multiple fading channels including AWGN, Rayleigh, Rician, Nakagami-m, Weibull, Suzuki, and Hoyt. The evaluation metrics include Bit Error Rate (BER) and spectral efficiency under varying Signal-to-Noise Ratio (SNR) and Doppler conditions.

1.2 Limitations of Existing Models

Existing research on OTFS is largely focused on mathematical modelling and theoretical derivation. While these studies establish the advantages of OTFS, they often:

- a) Consider only simplified channel models such as AWGN or Rayleigh
- b) Do not evaluate performance across multiple realistic fading conditions
- c) Lack simulation-based validation
- d) Do not analyse behaviour under diverse Doppler scenarios

Similarly, OFDM-based systems:

- a) Assume quasi-static channels
- b) Are highly sensitive to Doppler shifts
- c) Suffer from Inter-Carrier Interference (ICI)
- d) Show degraded performance in high mobility environments

Thus, there is a need for a comprehensive comparative analysis using realistic channel conditions.

1.3 Problem Statement

Modern communication systems face significant challenges due to time-varying channels, multipath propagation, and Doppler effects. OFDM, although efficient, fails to maintain performance under such conditions due to loss of subcarrier orthogonality, leading to increased BER and reduced efficiency.

Additionally, wireless channels vary significantly depending on environmental conditions and can be modelled using different statistical fading models such as Rayleigh, Rician, Nakagami-m, Weibull, Suzuki, and Hoyt distributions. However, most existing OTFS studies do not evaluate performance across such diverse channel models.

Therefore, the problem addressed in this project is to analyse and compare the performance of OFDM and OTFS under multiple fading channels and Doppler conditions, and to determine the effectiveness of OTFS in improving communication reliability and efficiency in high mobility environments.

2. LITERATURE SURVEY

2.1 Existing Methods

Orthogonal Frequency Division Multiplexing (OFDM) is the dominant modulation technique used in modern wireless communication systems such as LTE and 5G. It offers advantages such as high spectral efficiency, robustness to frequency-selective fading, and efficient implementation using Fast Fourier Transform (FFT).

However, OFDM operates in the time-frequency domain and assumes that the channel remains constant over the duration of a symbol. This assumption fails in high mobility environments, where the channel changes rapidly.

To address this limitation, Orthogonal Time Frequency Space (OTFS) modulation has been introduced. OTFS transforms information symbols into the delay-Doppler domain, where the channel appears more stable. This allows better exploitation of channel diversity and improved performance under time-varying conditions.

2.2 Evolution of OTFS and Its Advantages in High Mobility Scenarios

The rapid transition from 5G to next-generation 6G networks has highlighted the critical need for robust communication in high-mobility scenarios, such as high-speed railways, autonomous vehicles, and satellite communications. Conventional Orthogonal Frequency Division Multiplexing (OFDM), while effective in static environments, suffers significant performance degradation in these dynamic settings due to its sensitivity to Doppler spread. High relative velocities destroy the orthogonality of subcarriers, leading to severe Inter-Carrier Interference (ICI). In response, Orthogonal Time Frequency Space (OTFS) modulation has emerged as a superior alternative by multiplexing information symbols in the two-dimensional delay-Doppler (DD) domain. This approach leverages the stability and sparsity of the DD domain, where time-varying fading channels are transformed into a near-stationary representation. Consequently, OTFS provides full time-frequency diversity and is significantly more resilient to Doppler-induced impairments than traditional one-dimensional time or frequency domain schemes. Research into the practical implementation of OTFS has

focused heavily on developing efficient detection and equalization strategies to mitigate remaining interferences, such as Inter-Doppler Interference (IDI) caused by fractional Doppler shifts. While standard linear equalizers like Zero-Forcing (ZF) and Minimum Mean Square Error (MMSE) serve as foundational benchmarks, studies indicate that they experience a sharp decline in accuracy when fractional Doppler effects are present. To address this, sophisticated iterative algorithms like the Message Passing (MP) detector have been proposed, which exploit the sparsity of the delay-Doppler domain to refine symbol estimations and outperform traditional linear methods by an order of magnitude. Furthermore, recent advancements have explored integrating OTFS with Multiple-Input Multiple-Output (MIMO) architectures and diversity-combining techniques such as Equal-Gain Combining (EGC). These hybrid approaches have been shown to significantly reduce Bit Error Rates (BER) and enhance spectral efficiency, validating OTFS as a critical enabling technology for ultra-reliable, low-latency communication in the 6G era.

2.3 Advantages of Proposed Work

The proposed system addresses these limitations by:

- Evaluating performance across seven different fading channels
- Incorporating Doppler effects to simulate high mobility
- Providing comprehensive simulation results
- Comparing OFDM and OTFS under identical conditions

3. METHODOLOGY

3.1 System Architecture

The proposed system consists of a complete digital communication chain designed to evaluate the performance of OFDM and OTFS under various channel conditions. The system begins with random bit generation, followed by modulation, transmission through different channel models, and finally reception and error analysis.

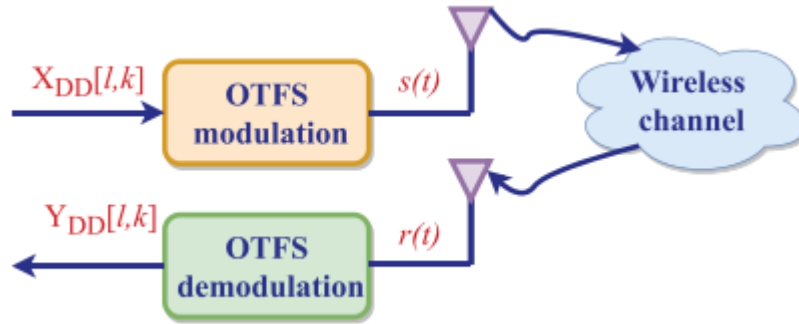


Figure 1: OTFS System Model

The major blocks of the system include:

- a) Random bit generator
- b) Modulation block (QAM/BPSK)
- c) OFDM/OTFS transmitter
- d) Channel model (fading + noise + Doppler)
- e) Receiver (equalization + demodulation)
- f) BER computation

In the OFDM system, data is transmitted across multiple orthogonal subcarriers, whereas in OTFS, data is mapped into the delay-Doppler domain before transmission.

3.2 System Design

3.2.1 OFDM System

In OFDM, the input data bits are first mapped into complex symbols using modulation schemes such as QAM. These symbols are then assigned to different subcarriers in the frequency domain.

The time-domain OFDM signal is obtained using the Inverse Fast Fourier Transform (IFFT):

$$s[n] = \frac{1}{M} \sum_{k=0}^{M-1} X[k] e^{j2\pi kn/M}$$

where M is the number of subcarriers and $X[k]$ represents the modulated data symbols.

To reduce inter-symbol interference caused by multipath propagation, a cyclic prefix (CP) is added to each OFDM symbol. At the receiver, the cyclic prefix is removed, and FFT is applied to recover the transmitted data.

However, in the presence of Doppler shift, the orthogonality between subcarriers is disturbed, leading to Inter-Carrier Interference (ICI).

3.2.2 OTFS System

OTFS is a two-dimensional modulation scheme that operates in the delay-Doppler domain. Unlike OFDM, which transmits symbols in the time-frequency domain, OTFS maps symbols onto a grid defined by delay and Doppler coordinates.

The transformation from delay-Doppler domain to time-frequency domain is performed using the Inverse Symplectic Finite Fourier Transform (ISFFT):

$$X_{TF}[m, n] = \frac{1}{\sqrt{MN}} \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} X_{DD}[l, k] e^{j2\pi(\frac{k}{N}m - \frac{l}{M}n)}$$

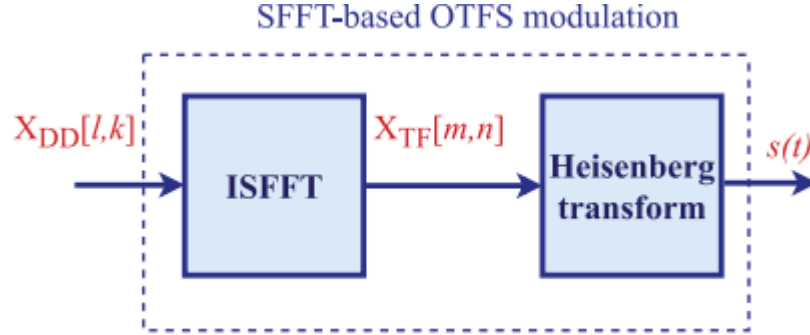


Figure 2: ISFFT-Based OTFS Modulation Block Diagram

At the receiver, the reverse transformation is performed using the Symplectic Finite Fourier Transform (SFFT):

$$Y_{DD}[l, k] = \frac{1}{\sqrt{MN}} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} Y_{TF}[m, n] e^{j2\pi(-\frac{m}{M}k + \frac{n}{N}l)}$$

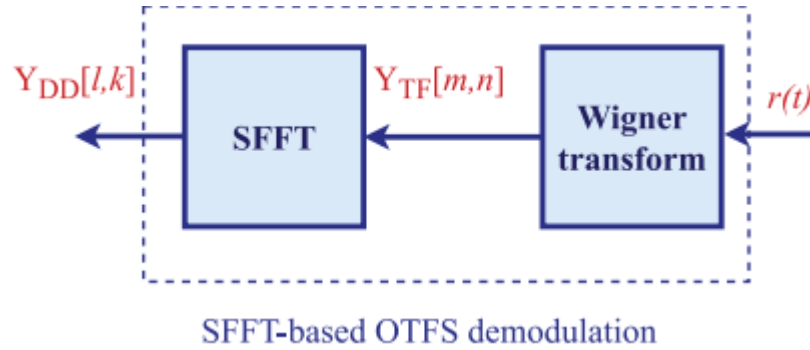


Figure 3: SFFT-Based OTFS demodulation Block Diagram

This representation allows OTFS to capture the entire channel characteristics and provides robustness against Doppler effects.

3.2.3 Channel Models

In this project, multiple fading channel models are considered to simulate real-world wireless environments.

- a) **AWGN Channel:** Adds Gaussian noise to the signal without fading. It has no fading, only noise

$$p(n) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-n^2/(2\sigma^2)}$$

- b) **Rayleigh Fading:** Models non-line-of-sight (NLOS) propagation with multiple scattered paths.

$$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2}{2\sigma^2}}, r \geq 0$$

- c) **Rician Fading:** Includes a dominant line-of-sight (LOS) component along with scattered signals in multipath.

$$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2+A^2}{2\sigma^2}} I_0\left(\frac{rA}{\sigma^2}\right), r \geq 0$$

- d) **Nakagami-m Fading:** A flexible model that can represent various fading conditions by adjusting parameter m .

$$p(r) = \frac{2m^m}{\Gamma(m)\Omega^m} r^{2m-1} e^{-\frac{mr^2}{\Omega}}, r \geq 0, m \geq \frac{1}{2}, \Omega = E[r^2]$$

- e) **Weibull Fading:** Used for modelling non-linear and complex propagation environments.

$$p(r) = \frac{k}{\lambda} \left(\frac{r}{\lambda}\right)^{k-1} e^{-(r/\lambda)^k}, r \geq 0$$

- f) **Suzuki Channel:** Combines Rayleigh fading with log-normal shadowing.

$$r = r_{\text{Rayleigh}} \times r_{\text{Lognormal}}$$

g) **Hoyt Channel:** Models asymmetric fading conditions.

$$p(r) = \frac{2q}{(1+q^2)\Omega} r \exp\left(-\frac{q^2 r^2}{(1+q^2)\Omega}\right) I_0\left(\frac{(1-q^2)r^2}{(1+q^2)\Omega}\right), 0 < q \leq 1$$

The general multipath channel model is given by:

$$r(t) = \sum_{i=1}^P h_i s(t - \tau_i) e^{j2\pi v_i(t - \tau_i)} + z(t)$$

Final BER Ordering

$$P_b(\text{Suzuki}) > P_b(\text{Rayleigh}) > P_b(\text{Hoyt}) > P_b(\text{Weibull}) > P_b(\text{Rician}) \\ > P_b(\text{Nakagami } (m > 1)) > P_b(\text{AWGN})$$

3.2.4 Doppler Effect Consideration

Doppler shift occurs due to relative motion between transmitter and receiver. It results in a frequency shift in the received signal.

The Doppler frequency is given by:

$$f_d = \frac{vf_c}{c}$$

The Doppler effect is introduced in the signal as:

$$e^{j2\pi f_d t}$$

This effect causes time variation in the channel, which significantly affects OFDM performance but is effectively handled by OTFS.

3.2.5 Delay-Doppler Domain

The delay-Doppler domain represents the channel in terms of time delay and frequency shift. Each path in the channel is represented as a point in this domain.

The channel in delay-Doppler domain is expressed as:

$$h(\tau, \nu) = \sum h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i)$$

This representation makes OTFS more suitable for high mobility environments, as the channel appears more stable compared to time-frequency representation.

3.3 Working Principle

The system operates as follows:

1. Random binary data is generated.
2. Data is modulated using QAM/BPSK.
3. The signal is transmitted using OFDM and OTFS systems.
4. The transmitted signal passes through different fading channels.
5. Noise and Doppler effects are added.
6. At the receiver, equalization is performed.
7. The signal is demodulated and BER is calculated.

4. IMPLEMENTATION

4.1 Simulation Setup

The system is implemented using MATLAB simulation with the following parameters:

- Number of subcarriers (M) = 64
- Number of symbols (N) = 16
- Modulation scheme = 16-QAM
- SNR range = 0 to 30 dB (step size of 5 dB)
- Doppler frequency range = 0 to 1000 Hz (step size of 100 Hz)
- Number of frames = 50
- Cyclic prefix size = 16

4.2 Algorithms Used

The following algorithms are used:

a) Fast Fourier Transform (FFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

Converts a signal from time domain $\rightarrow$ frequency domain. Reduces complexity from $O(N^2)$ to $O(N \log N)$. In communication: used in OFDM/OTFS receivers to analyze subcarriers

b) Inverse FFT (IFFT)

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$$

Converts frequency domain $\rightarrow$ time domain. Used at transmitter to generate time-domain signal from symbols. Essential for multicarrier systems (OFDM, OTFS)

c) ISFFT and SFFT (for OTFS)

ISFFT (Inverse Symplectic FFT): Used in OTFS modulation

$$X[k, l]_{\text{Delay-Doppler}} \rightarrow X[n, m]_{\text{Time-Frequency}}$$

Maps symbols from delay-Doppler domain $\rightarrow$ time-frequency domain. Prepares signal for transmission

SFFT (Symplectic FFT): Reverse of ISFFT

$$X[n, m]_{\text{Time-Frequency}} \rightarrow X[k, l]_{\text{Delay-Doppler}}$$

Used at receiver in OTFS. Recovers symbols in delay-Doppler domain (more robust to fading)

d) Zero-Forcing Equalization

Removes channel distortion by inverting channel effect

$$\hat{X} = \frac{Y}{H}$$

Y : received signal, H : channel response. Forces interference to zero

e) Bit Error Rate calculation

Measures system performance

$$\text{BER} = \frac{\text{Number of error bits}}{\text{Total transmitted bits}}$$

Lower BER $\Rightarrow$ better system. Used to compare channels (Rayleigh, Rician, etc.)

4.3 Code Explanation

The MATLAB program simulates a complete communication system for both OFDM and OTFS. The execution follows a structured sequence of steps, which are repeated for different Signal-to-Noise Ratio (SNR) and Doppler values to evaluate performance.

Step 1: System Initialization

The simulation begins by defining key system parameters such as the number of subcarriers, number of symbols, modulation order (16-QAM), cyclic prefix length, number of frames, and ranges for SNR and Doppler frequency. These parameters control the size of transmitted data and the operating conditions of the system.

Step 2: Random Bit Generation

For each frame, a random sequence of binary bits is generated. These bits represent the information to be transmitted. The total number of bits depends on the number of subcarriers, symbols, and modulation scheme used.

Step 3: Symbol Modulation

The generated binary data is converted into complex symbols using 16-QAM modulation. In this process, groups of bits are mapped to constellation points. The resulting symbols are then arranged into a matrix structure representing the transmission grid.

Step 4: OTFS Transmission Process

For the OTFS system, the modulated symbols are first interpreted in the delay-Doppler domain. These symbols are then transformed into the time-frequency domain using a combination of inverse Fourier transforms. After this transformation, the signal is converted into the time domain and a cyclic prefix is added to each symbol to reduce inter-symbol interference. Finally, the signal is converted into a serial stream for transmission.

Step 5: OFDM Transmission Process

In the OFDM system, the modulated symbols are directly mapped onto subcarriers in the frequency domain. An inverse Fourier transform is applied to convert the signal into the time domain. Similar to OTFS, a cyclic prefix is added to each symbol. The signal is then serialized for transmission.

Step 6: Channel Modelling

The transmitted signals are passed through different wireless channel models. These channels simulate real-world propagation conditions such as multipath fading. The channel introduces effects like signal attenuation and delay by combining multiple delayed and scaled versions of the transmitted signal.

Different types of fading channels are considered, including Rayleigh, Rician, Nakagami, and others, each representing a specific propagation environment.

Step 7: Noise Addition

Additive White Gaussian Noise (AWGN) is added to the transmitted signal based on the specified SNR value. This step simulates the effect of thermal noise present in real communication systems.

Step 8: Doppler Effect Application

To simulate high mobility scenarios, Doppler shift is introduced into the received signal. This causes a time-varying frequency shift, representing the effect of relative motion between the transmitter and receiver.

Step 9: Signal Reception and Equalization

At the receiver, the signal is first reshaped and the cyclic prefix is removed. The signal is then converted back into the frequency domain using Fourier transform operations.

Channel equalization is performed using a Zero-Forcing method, where the received signal is divided by the channel response to compensate for channel distortions.

Step 10: OTFS and OFDM Demodulation

For OTFS, additional transformations are applied to convert the signal back from the time-frequency domain to the delay-Doppler domain. For OFDM, the equalized frequency-domain symbols are directly used.

The symbols are then demodulated to recover the transmitted bits.

Step 11: BER Calculation

The recovered bits are compared with the originally transmitted bits to determine the number of errors. The Bit Error Rate (BER) is calculated as the ratio of erroneous bits to total transmitted bits.

Step 12: Iterative Simulation

The entire process is repeated for multiple frames and for different values of SNR and Doppler frequency. The BER values are averaged over all iterations to obtain accurate and reliable performance results.

5. RESULTS

5.1 BER vs SNR (OFDM)

The BER performance of OFDM improves as SNR increases. However, in fading channels such as Rayleigh and Nakagami, the BER remains relatively high due to multipath interference and Doppler effects.

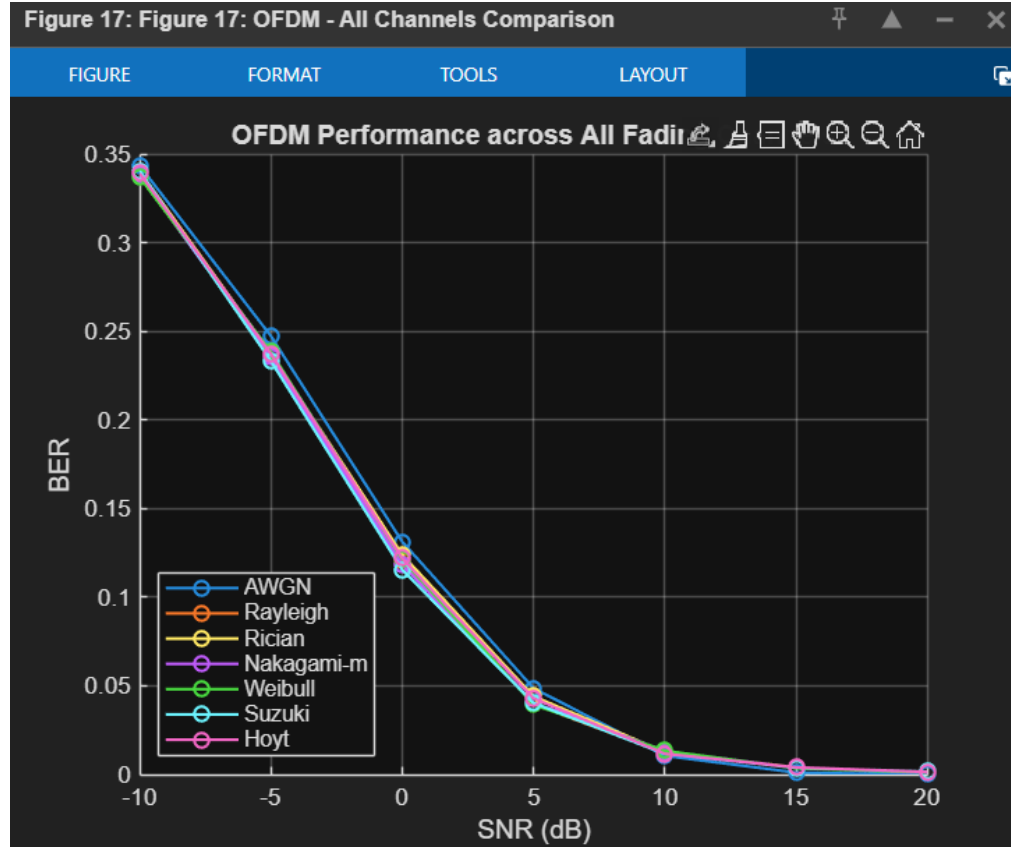


Figure 4: BER vs SNR Performance of OFDM under Different Fading Channels

5.2 BER vs SNR (OTFS)

OTFS shows significantly lower BER compared to OFDM across all channel models. This is due to its ability to spread symbols across the delay-Doppler domain, utilizing full channel diversity.

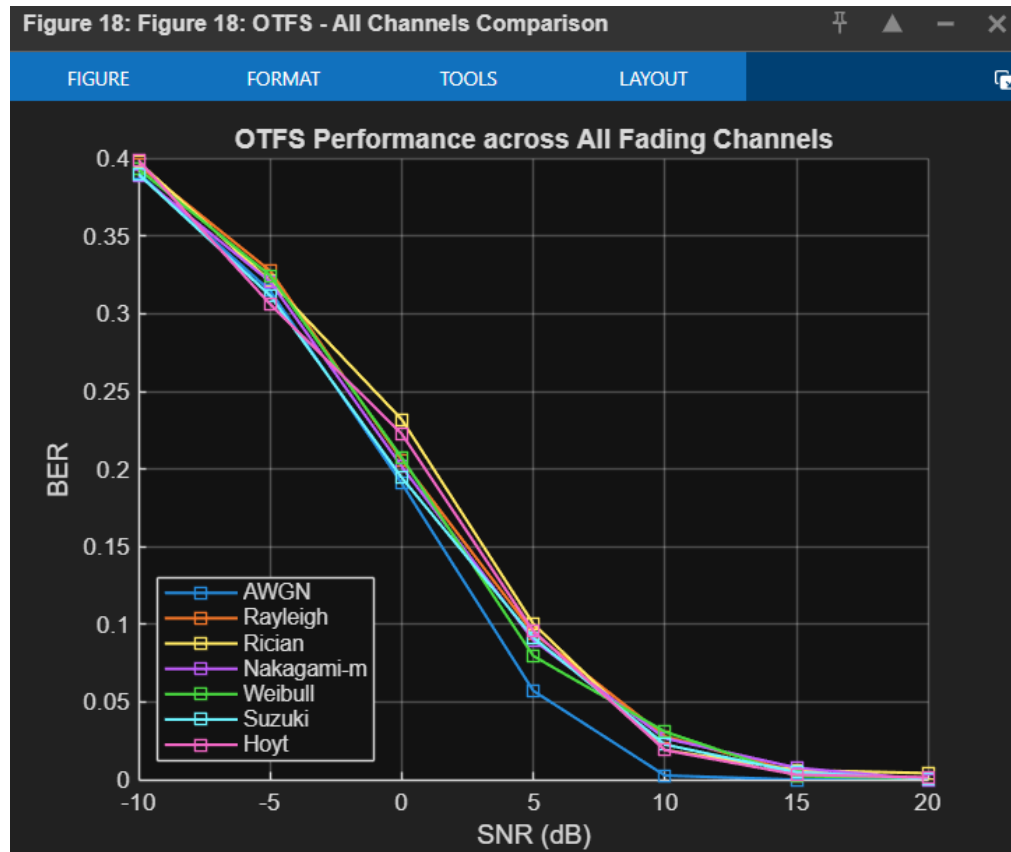


Figure 5: BER vs SNR Performance of OTFS under Different Fading Channels

5.3 Doppler Impact Analysis

As Doppler frequency increases:

- OFDM performance degrades rapidly due to ICI
- OTFS remains stable due to delay-Doppler representation

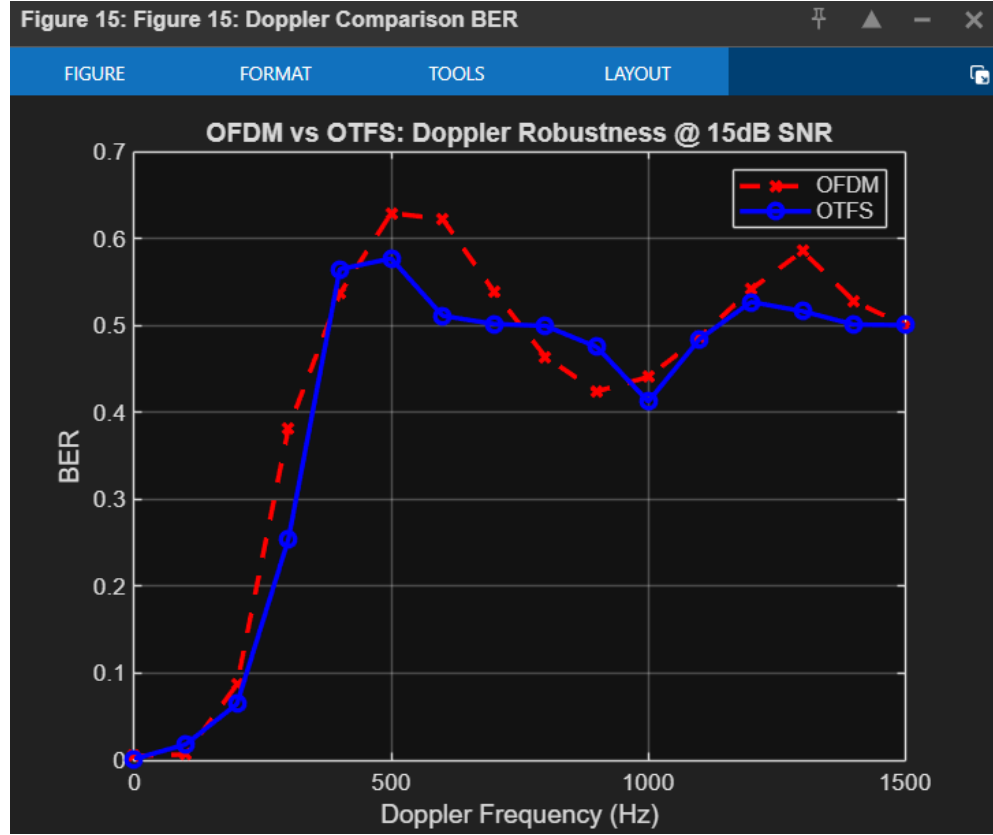


Figure 6: BER Comparison of OFDM and OTFS under Doppler Frequency Variation at 15 dB SNR

5.4 Spectral Efficiency Analysis

$$\eta = (1 - BER) \log_2(M)$$

Spectral efficiency decreases with increasing BER. OTFS maintains higher efficiency under high Doppler conditions.

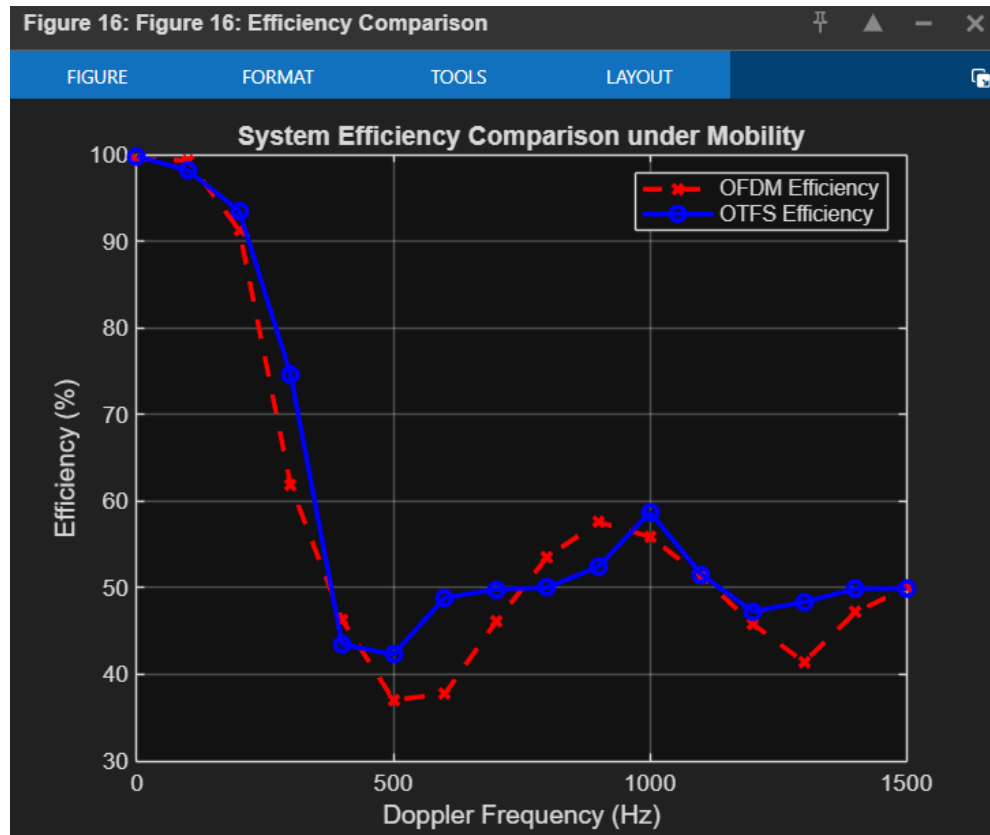


Figure 7: Spectral Efficiency Comparison of OFDM and OTFS under Doppler Frequency Variation

5.5 Performance Comparison

- a) OTFS provides lower BER in all channel conditions
- b) OFDM performance degrades under high mobility
- c) OTFS maintains better spectral efficiency
- d) OTFS is more robust to Doppler effects

6. CONCLUSION

6.1 Summary of Results

The simulation results clearly show the comparative performance of OFDM and OTFS under different channel conditions and Doppler effects. From the BER vs SNR analysis, it is observed that both systems improve as SNR increases; however, OTFS consistently achieves a lower BER than OFDM across all channel models.

In fading channels such as Rayleigh, Rician, and Nakagami, OFDM performance degrades due to multipath effects and loss of subcarrier orthogonality. In contrast, OTFS provides better reliability by utilizing the delay-Doppler representation of the channel.

The Doppler analysis indicates that OFDM is highly sensitive to frequency shifts, resulting in a rapid increase in BER as Doppler increases. OTFS, however, maintains stable performance under high Doppler conditions.

Similarly, spectral efficiency is higher for OTFS, especially in dynamic environments, due to its lower error rates. Overall, the results confirm that OTFS outperforms OFDM in terms of robustness, reliability, and efficiency.

6.2 Final Comparison (OTFS vs OFDM)

A comparison between OFDM and OTFS based on simulation results is summarized below:

Feature	OFDM	OTFS
BER Performance	Higher BER in fading channels	Lower BER across all channels
Doppler Handling	Poor	Excellent
Spectral Efficiency	Decreases at high Doppler	Remains relatively stable
Channel Robustness	Sensitive to variations	Highly robust
Suitability	Low mobility scenarios	High mobility scenarios

From the comparison, it is evident that OTFS provides significant improvements over OFDM, particularly in environments with high mobility and complex channel conditions.

7. FUTURE SCOPE

The current work can be further improved by incorporating higher-order modulation schemes such as 64-QAM or 256-QAM to achieve higher data rates, and by implementing advanced equalization techniques like MMSE (Minimum Mean Square Error) to enhance system performance and reduce noise amplification compared to Zero-Forcing. In addition, the system can be extended to real-time implementation using hardware platforms such as FPGA or Software Defined Radio (SDR), enabling practical validation of simulation results and performance evaluation under real-world channel conditions. Furthermore, the Orthogonal Time Frequency Space modulation (OTFS) technique has strong potential for next-generation communication systems such as 6G, where high mobility, low latency, and reliability are essential, making it highly suitable for applications including autonomous vehicles, high-speed rail communication, and satellite communication systems.

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